Energy 168 (2019) 1157–1167

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Effect of acetone-butanol-ethanol (ABE)–gasoline blends on regulated and unregulated emissions in spark-ignition engine

Yuanxu Li [a, b], Zhi Ning [a, *], Chia-fon F. Lee [b, c, **], Junhao Yan [b], Timothy H. Lee [b]

[a] School of Mechanical, Electronic and Control Engineering, Beijing Jiaotong University, Beijing 100044, China
[b] Department of Mechanical Science and Engineering, University of Illinois at Urbana-Champaign, IL 61801, USA
[c] School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China

ARTICLE INFO

Article history:
Received 6 April 2018
Received in revised form
27 July 2018
Accepted 4 December 2018
Available online 5 December 2018

Keywords:
Acetone-butanol-ethanol
Gas chromatography
Unregulated emissions
SI engine
Alternative fuel

ABSTRACT

An experimental study was conducted in a port-fuel injection (PFI) spark-ignition (SI) engine fueled with acetone-butanol-ethanol (ABE) and gasoline. 30 vol% ABE blends with different component ratios (A:B:E = 3:6:1, 6:3:1, and 5:14:1), referred to as ABE(3:6:1)30, ABE(6:3:1)30, and ABE(5:14:1)30, were used as test fuels. Additionally, 30 vol% ethanol and 30% vol.% butanol blended with 70 vol% gasoline, referred to as E30 and B30, respectively, were also used as test fuels. Both the regulated emissions, which include unburned hydrocarbons (UHC), carbon monoxide (CO), and nitrogen oxide (NO_x), and unregulated emissions, including acetaldehyde, 1,3-butadiene, benzene, toluene, ethylbenzene, and xylene (BTEX), were investigated under various conditions. The experiments were conducted at an engine speed of 1200 rpm and at engine loads of 3, 4, 5, and 6 bar brake mean effective pressure (BMEP) under various equivalence ratios ($\varphi = 0.83–1.25$). The results indicated that ABE(6:3:1)30 had the lowest UHC and CO emissions, with ABE(3:6:1)30 having slightly higher NO_x emissions. As for unregulated emissions, B30 showed the highest acetaldehyde emission and ABE(6:3:1)30 had the lowest among all the fuel blends. ABE(3:6:1)30 and ABE(6:3:1)30 showed a reduction in 1,3-butadiene emission. Regarding the BTEX emissions, an overall reduction was observed for all the fuel blends, with ABE(3:6:1)30 having shown the lowest BTEX emissions among all the test fuels. The results indicate that ABE could be used as a promising alternative fuel in SI engines, owing to the reduction in emissions.

1. Introduction

In recent years, butanol has received increasing attention as a promising renewable alternative fuel for internal combustion engines [1,2]. Compared to ethanol, which is the most commonly used alternative fuels, butanol presents higher heating value, lower volatility, lower ignition problems, higher viscosity, and higher lubricity [3–5]. Currently, there are two main methods for the production of butanol: one method is to produce it from fossil fuels (petro-butanol) and the other method is to produce it from biomass (bio-butanol). Petro-butanol is usually synthetic and derived from a petrochemical reaction [6]. However, the costs of petro-butanol production are closely linked to the price of crude oil and petro-butanol is highly dependent on petroleum derivatives [7]. As a result, this type of method is not the best solution to relieve the energy crisis and at the same time, is not environmentally friendly. Bio-butanol can be produced from the fermentation of carbohydrates, thereby converted it from sugarcane, bagasse, and other non-food feedstock [3,8]. Because bio-butanol's intermediate products are acetone, butanol, and ethanol, this process is also known as ABE fermentation. The lower cost of fermentation production results in a much cheaper price of bio-butanol (US$0.83/L) than that of petro-butanol (US$1.34/L) produced by chemical synthesis [9]. In addition, producing butanol by ABE fermentation is in line with the requirements for utilizing renewable energy sources and stringent pollution regulations. China, which is one of the largest producers of bio-butanol, has already increased the total ABE production to 210,000 tons and it is expected to reach 1,000,000 tons in the coming years [10,11].

Unfortunately, using the ABE fermentation process to produce butanol still has some limitations: low butanol production yield increases the usage of feedstock and the final costs, and the recovery

* Corresponding author.
** Corresponding author. Department of Mechanical Science and Engineering, University of Illinois at Urbana-Champaign, IL, 61801, USA.
E-mail addresses: zhining@bjtu.edu.cn (Z. Ning), cflee@illinois.edu (C.-f.F. Lee).

https://doi.org/10.1016/j.energy.2018.12.022
0360-5442/